

Diet, lung function, and lung function decline in a cohort of 2512 middle aged men



BACKGROUND—A prospective cohort study of 2512 Welshmen aged 45-59 living in Caerphilly in 1979-1983 was used to investigate associations between diet and lung function. **METHODS**—At baseline (phase I) and at five year follow up (phase II), forced expiratory volume in one second (FEV1) was measured using a McDermott spirometer and dietary data were obtained using a semi-quantitative food frequency questionnaire. **RESULTS**—Good lung function, indicated by high maximum FEV1 given age and height, was associated with high intakes of vitamin C, vitamin E, β -carotene, citrus fruit, apples, and the frequent consumption of fruit juices/squashes. Lung function was inversely associated with magnesium intake but there was no evidence of an association with fatty fish. Following adjustment for confounders including body mass index, smoking history, social class, exercise, and total energy intake, only the associations with vitamin E and apples persisted, with lung function estimated to be 39 ml (95% confidence interval (CI) 9 to 69) higher for vitamin E intakes one standard deviation (SD) apart and 138 ml higher (95% CI 58 to 218) for those eating five or more apples per week compared with non-consumers. Decline in lung function between phases was not significantly associated with the changing intakes of apples or vitamin E. An association between high average apple consumption and slow decline in lung function lost significance after adjustment for confounders. **CONCLUSIONS**—A strong positive association is seen between lung function and the number of apples eaten per week cross sectionally, consistent with a protective effect of hard fruit rather than soft/citrus fruit. The recent suggestion that such effects are reversible was not supported by our longitudinal analysis.